Aadarsh Sahoo

Ph.D. Student, Kortschak Scholar
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Education and Positions

California Institute of Technology, Pasadena, USA — Sep 2023 - June 2028
Computing and Mathematical Sciences, Ph.D. Student (Kortschak Scholar) — Advisor: Prof. Georgia Gkioxari

MIT-IBM Watson AI Lab, Cambridge, USA — July 2022 - Aug 2023
Pre-doctoral Student Researcher

Indian Institute of Technology Kharagpur, India — July 2017 - Apr 2022
Major in Computer Science & Engineering, Dual Degree — GPA **9.41/10.00**

Interests

My research focuses on building multimodal foundation models that perceive, reason, and interact in 3D environments. I develop unified representations bridging geometry, vision, and language through 3D tokenization, conversational visual understanding, and parameter-efficient model adaptation. My goal is to advance embodied AI systems that robustly interpret and act within real-world environments to benefit society.

Publications

Aadarsh Sahoo, Vansh Tibrewal, Georgia Gkioxari. *Aligning Text, Images, and 3D Structure Token-by-Token.* Under peer-review.

Omprakash Chakraborty, Aadarsh Sahoo, Rameswar Panda, Abir Das. *XPL: A Cross-Model framework for Semi-Supervised Prompt Learning in Vision-Language Models.*
Accepted at **TMLR 06/2024**.

Omprakash Chakraborty, Aadarsh Sahoo, Rameswar Panda, Abir Das. *AnyDA: Anytime Domain Adaptation.*
Accepted at **ICLR 2023**.

Aadarsh Sahoo, Rutav Shah, Rameswar Panda, Kate Saenko, Abir Das. *Contrast and Mix: Temporal Contrastive Video Domain Adaptation with Background Mixing.*
Accepted at **NeurIPS 2021**.

Aadarsh Sahoo, Rameswar Panda, Rogerio Feris, Kate Saenko, Abir Das. *Select, Label, and Mix: Learning Discriminative Invariant Feature Representations for Partial Domain Adaptation.*
Best Paper Honorable Mention at **WACV 2023**.

Aadarsh Sahoo, Ankit Singh, Rameswar Panda, Rogerio Feris, Abir Das. *Mitigating Dataset Imbalance via Joint Generation and Classification.*
Accepted at **IPCV Workshop at ECCV 2020**.

Aadarsh Sahoo, Anshuman Senapati, Abir Das, Yoon Kim, Rogerio Feris, Rameswar Panda. *Frustratingly Simple Contrastive Prompt Tuning for Vision-Language Models*

Research Experience

Glab, Caltech – Conversational Multi-Modal Large Language Models — July 2025 - Present
Advisors: Prof. Georgia Gkioxari, Prof. Pietro Perona

- Developing conversational image segmentation models for conversational natural language queries.
- Building visual reasoning systems that handle relational understanding, comparative attributes, counterfactual reasoning, and conditional logic to segment objects beyond simple labels (e.g., "what can I use to prop a door open?" or "which objects are most likely to fall?").
- Investigating whether grounding visual representations in conversational concepts enhances multi-modal understanding and enables more robust robotics applications.

Glab, Caltech – 3D Multi-Modal Large Language Models Oct 2023 - Present
Advisors: Prof. Georgia Gkioxari, Prof. Pietro Perona

- Researching 3D perception and reasoning with Large Multi-Modal Models, focusing on grounded alignment between visual and language modalities.
- Designed a unified representation space integrating 3D geometry, images, and text to enable sequential modeling across modalities.
- Developed novel 3D tokenization using vector-quantization techniques to reframe complex tasks like rendering as sequence prediction problems, enabling language model architectures for structured 3D representations.

MIT-IBM Watson AI Lab – Pretraining and Finetuning Large Language Models Feb 2023 - July 2023
Advisor: Dr. Rameswar Panda

- Was part of IBM's Granite LLM team responsible for engineering and pretraining LLMs (3B–13B) for enterprise.
- Developed a 3B finance model surpassing BloombergGPT; evaluated on domain-specific and general benchmarks.
- Incorporated parameter-efficient finetuning; work featured in the IBM CEO's keynote at Think 2023.

MIT-IBM Watson AI Lab – Prompt Tuning Vision-Language Models; DARPA LwLL July 2022 - Jan 2023
Advisors: Dr. Rameswar Panda, Dr. Rogerio Feris, Prof. Yoon Kim

- Proposed a relational contrastive learning method (CPT) for cross-modal consistency & improved generalization.
- Achieved SOTA across 15 datasets and 10 vision-language backbones (PyTorch).
- CPT underpins a multi-billion-dollar FMaaS product; contributed to DARPA LwLL evaluations.

CVIR IIT Kharagpur – Anytime Domain Adaptation Jan 2022 - May 2022
Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)

- Developed a single network adaptable to various computational budgets for domain adaptation.
- Proposed a recursive distillation strategy for robust low-budget networks; published at ICLR 2023.

UC Berkeley and Boston University – DARPA LwLL May 2021 - Aug 2021
Advisors: Prof. Trevor Darrell, Prof. Kate Saenko

- Contributed to few-shot action recognition for video classification under the DARPA LwLL project.
- Proposed a spatio-temporal contrastive learning approach for temporal/spatial invariance.

CVIR IIT Kharagpur – Unsupervised Domain Adaptation in Videos Jan 2021 - May 2021
Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)

- Achieved SOTA in action recognition adaptation via contrastive learning and background mixing.
- Leveraged video speed invariance and target self-supervision; published at NeurIPS 2021.

CVIR IIT Kharagpur – Dynamic Source Selection for Partial Domain Adaptation May 2020 - Oct 2020
Advisors: Prof. Abir Das (IIT Kharagpur), Dr. Rameswar Panda (MIT-IBM Watson AI Lab)

- Devised a dynamic source selection, outlier detection, and mix-up approach for partial domain adaptation.
- Incorporated Gumbel-Softmax-based policy networks; published at WACV 2023.

Activities

Reviewer. CVPR, NeurIPS, ICML, ECCV, ICCV, ICLR, WACV, TPAMI
Student Volunteer and Reviewer. CVPR-2021/22 workshop on Dynamic Neural Networks Meets Computer Vision.
Teaching Assistant. CS60010 Deep Learning, CS60077 Reinforcement Learning, IIT Kharagpur.
Mentor. Mentored multiple undergrad students for research.

Honors & Awards

Outstanding Reviewer. ECCV 2024
Outstanding Reviewer. CVPR 2024
Awarded Kortschak Scholarship. Computing and Mathematical Sciences, Caltech
Best Paper Honorable Mention Award for Select, Label, and Mix paper. WACV 2023
Received an offer for Pre-doctoral researcher. Allen Institute for AI, Seattle
Secured a position in the Top-10; Granted Department Change, 2018. IIT Kharagpur, India
All India Joint Entrance Examination, 2017. Joint Seat Allocation Authority (JoSAA), India
Regional Mathematics Olympiad (RMO), 2014. Homi Bhabha Centre for Science Education (HBCSE)

Programming Skills

Languages: Python, C, C++, Matlab, LaTeX
Deep Learning: PyTorch, TensorFlow, Keras

Relevant Coursework

Probability and Statistics, Linear Algebra, Machine Learning, Deep Learning, 3D Deep Learning, Large Language and Vision Models, Reinforcement Learning, Artificial Intelligence, Natural Language Processing, Scalable Data Mining, Algorithms, Formal Language and Automata Theory, Operating Systems, Computer Networks, Compilers, Computer Architecture, Discrete Structures.
Online MOOCs: CS231n: CNNs for Visual Recognition by Stanford, CS294-158: Deep Unsupervised Learning by UC Berkeley, Deep Learning Specialization by deeplearning.ai, Machine Learning by Stanford-Online.